HARLEEN KAUR BHULLAR

harleenbhullar0@gmail.com — 9463408797 — LinkedIn — GitHub

EDUCATION

Bachelor of Technology in Computer Science and Engineering Aug 2023 – 2027
Guru Nanak Dev Engineering College, Ludhiana CGPA: 9.03/10

EXPERIENCE

Research Intern, DRDO, Delhi Jun 2026 – Jul 2026

- Built and engineered an end-to-end multi-modal data pipeline in Python, implementing a few-shot CLIP and BART-based classification architecture to classify complex organization-specific textual and visual content.
- Evaluated pipeline performance on manually annotated datasets, achieving 52% accuracy for multi-label text and 68% accuracy for multi-class image classification while identifying alignment divergence between modalities.

Open Source Contributor Present

- Contributed full-stack updates to JavaScript (React, Node.js) and Python projects during Hacktoberfest, GSSoC, and SWoC, collaborating with remote teams via Git version control workflows.
- Developed modular UI components, created new application pages, and resolved page linking and routing configurations to enhance frontend navigation.
- Built a physics-based water simulation mode entirely in JavaScript for the Sugarizer repository to model custom object behavior under water.

Intern, I3C & IIT Jammu, GNDEC Jun 2025 – Aug 2025

- Programmed automated data analysis scripts using Python libraries including Pandas, NumPy, and Matplotlib to run exploratory analysis and visualization on structured customer and sports datasets.
- Developed and evaluated machine learning backend models for customer churn prediction, optimizing algorithmic workflows to increase macro F1-score from 51% to 77%.

PROJECTS

VoiceBridge (Full-Stack Voice-to-Voice Translation) *React, Node.js, Socket.IO, Vercel*
Developed a real-time full-stack multilingual communication platform utilizing React, Node.js, and Socket.IO for seamless voice interactions. Implemented real-time speech-to-text, translation via Facebook NLLB-200, and audio synthesis, deploying the frontend architecture on Vercel.
Live Demo: voice-bridge-zeta.vercel.app

Raunaq (Punjabi Movie Platform) *React, Python, Flask, NLP*
Built a full-stack movie recommendation and discovery application featuring a React frontend and a Python Flask API backend. Engineered a custom content-based filtering system using Natural Language Processing to process movie indices and serve dynamic listings.
Live Demo: raunaqpunjabimovie.vercel.app

PrepMate (Voice-Driven Interview Simulator) *LangChain, LLM APIs, RAG*
Developed a full-stack voice-based interview simulator using LangChain and large language model APIs. Engineered a Retrieval-Augmented Generation (RAG) backend architecture to parse structured PDF resumes and job descriptions for real-time evaluation sessions.
GitHub: github.com/HarleenKaurBhullar/PrepMate

Harmful Content Detection & Extension *Python, CLIP, BERT*
Engineered a multimodal content detection application utilizing integrated CLIP and BERT models. Developed a demonstration website alongside a mock browser extension to showcase real-time web safety content screening.
GitHub: github.com/HarleenKaurBhullar/HarmfulMemeDetection

TRAINING

Prompt Engineering Jun 2025 – Jul 2025
Guru Nanak Dev Engineering College, Ludhiana
Completed specialized course framework focused on context engineering, designing effective system prompts, and orchestrating multi-agent architectures using LangChain and programmatic APIs.

KEY SKILLS

Languages: JavaScript, Python, C++
Frameworks/Libraries: React, Node.js, Express, FastAPI, Flask, PyTorch, LangChain
Tools/Databases: Git, GitHub, MongoDB, Vercel